



Syllabus: Matrices, determinants, System of linear equations, Eigenvalues and eigenvectors, LU decomposition.

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	1	2	1	0	2	1	0	1	0	1	2
2 Marks Count	1	1	1	1	0	2	1	1	0	1	2
Total Marks	3	4	3	2	2	5	2	3	2	3	5

6.1 Cartesian Coordinates (1)

6.1.1 Cartesian Coordinates: GATE IT 2007 | Question: 80



Let $P_1, P_2, \dots, P_n$ be n points in the xy -plane such that no three of them are collinear. For every pair of points P_i and P_j , let L_{ij} be the line passing through them. Let L_{ab} be the line with the steepest gradient amongst all $\frac{n(n-1)}{2}$ lines.

Which one of the following properties should necessarily be satisfied ?

- A. P_a and P_b are adjacent to each other with respect to their x -coordinate
- B. Either P_a or P_b has the largest or the smallest y -coordinate among all the points
- C. The difference between x -coordinates P_a and P_b is minimum
- D. None of the above

gateit-2007 linear-algebra cartesian-coordinates

Answer key

6.2 Determinant (11)

6.2.1 Determinant: GATE CSE 1997 | Question: 1.3



The determinant of the matrix $\begin{bmatrix} 6 & -8 & 1 & 1 \\ 0 & 2 & 4 & 6 \\ 0 & 0 & 4 & 8 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

- A. 11
- B. -48
- C. 0
- D. -24

gate1997 linear-algebra normal determinant

Answer key

6.2.2 Determinant: GATE CSE 2000 | Question: 1.3



The determinant of the matrix

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 8 & 1 & 7 & 2 \\ 2 & 0 & 2 & 0 \\ 9 & 0 & 6 & 1 \end{bmatrix}$$

- A. 4
- B. 0
- C. 15
- D. 20

gatecse-2000 linear-algebra easy determinant

Answer key

6.2.3 Determinant: GATE CSE 2013 | Question: 3



Which one of the following does NOT equal

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} ?$$

A. $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$

C. $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$

B. $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

D. $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

gatecse-2013 linear-algebra normal determinant

Answer key

6.2.4 Determinant: GATE CSE 2014 Set 2 | Question: 4



If the matrix A is such that

$$A = \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} [1 \ 9 \ 5]$$

then the determinant of A is equal to _____.

gatecse-2014-set2 linear-algebra numerical-answers easy determinant

Answer key

6.2.5 Determinant: GATE CSE 2019 | Question: 9



Let X be a square matrix. Consider the following two statements on X .

- I. X is invertible
- II. Determinant of X is non-zero

Which one of the following is TRUE?

- A. I implies II; II does not imply I
- B. II implies I; I does not imply II
- C. I does not imply II; II does not imply I
- D. I and II are equivalent statements

gatecse-2019 engineering-mathematics linear-algebra determinant one-mark

Answer key

6.2.6 Determinant: GATE CSE 2023 | Question: 8



Let

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B , respectively.

Which one of the options given below is TRUE?

- A. $\det(A) = \det(B)$
- B. $\det(B) = -\det(A)$
- C. $\det(A) = 0$
- D. $\det(AB) = \det(A) + \det(B)$

gatecse-2023 linear-algebra determinant one-mark easy

Answer key

6.2.7 Determinant: GATE CSE 2024 | Set 2 | Question: 37



Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- A. The determinant of B is the negative of the determinant of A
- B. If A is invertible, then B is also invertible
- C. If A is symmetric, then B is also symmetric
- D. If the trace of A is zero, then the trace of B is also zero

gatecse2024-set2 linear-algebra multiple-selects matrix determinant two-marks

Answer key

6.2.8 Determinant: GATE CSE 2025 | Set 2 | Question: 4



Let L , M , and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$. Which ONE of the following is the value of the determinant of $(M - N)$?

- A. 0
- B. 1
- C. 2
- D. 3

gatecse2025-set2 linear-algebra determinant easy one-mark

Answer key

6.2.9 Determinant: GATE DS&AI 2024 | Question: 25



Consider the 3×3 matrix $M = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 3 \\ 4 & 3 & 6 \end{bmatrix}$.

The determinant of $(M^2 + 12M)$ is _____.

gate-ds-ai-2024 numerical-answers matrix determinant linear-algebra easy one-mark

Answer key

6.2.10 Determinant: GATE IT 2004 | Question: 32



Let A be an $n \times n$ matrix of the following form.

$$A = \begin{bmatrix} 3 & 1 & 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & \dots & 0 & 0 & 0 \\ \dots & & & & & & & & \\ \dots & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \dots & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 & 1 & 3 \end{bmatrix}_{n \times n}$$

What is the value of the determinant of A ?

- A. $\left(\frac{5+\sqrt{3}}{2}\right)^{n-1} \left(\frac{5\sqrt{3}+7}{2\sqrt{3}}\right) + \left(\frac{5-\sqrt{3}}{2}\right)^{n-1} \left(\frac{5\sqrt{3}-7}{2\sqrt{3}}\right)$

- B. $\left(\frac{7+\sqrt{5}}{2}\right)^{n-1} \left(\frac{7\sqrt{5}+3}{2\sqrt{5}}\right) + \left(\frac{7-\sqrt{5}}{2}\right)^{n-1} \left(\frac{7\sqrt{5}-3}{2\sqrt{5}}\right)$
- C. $\left(\frac{3+\sqrt{7}}{2}\right)^{n-1} \left(\frac{3\sqrt{7}+5}{2\sqrt{7}}\right) + \left(\frac{3-\sqrt{7}}{2}\right)^{n-1} \left(\frac{3\sqrt{7}-5}{2\sqrt{7}}\right)$
- D. $\left(\frac{3+\sqrt{5}}{2}\right)^{n-1} \left(\frac{3\sqrt{5}+7}{2\sqrt{5}}\right) + \left(\frac{3-\sqrt{5}}{2}\right)^{n-1} \left(\frac{3\sqrt{5}-7}{2\sqrt{5}}\right)$

gateit-2004 linear-algebra matrix normal determinant

Answer key

6.2.11 Determinant: GATE IT 2005 | Question: 3



The determinant of the matrix given below is

$$\begin{bmatrix} 0 & 1 & 0 & 2 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

- A. -1 B. 0 C. 1 D. 2

gateit-2005 linear-algebra normal determinant

Answer key

6.3 Eigen Value (33)

6.3.1 Eigen Value: GATE CSE 1993 | Question: 01.1



The eigen vector (s) of the matrix

$$\begin{bmatrix} 0 & 0 & \alpha \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \alpha \neq 0$$

is (are)

- A. $(0,0,\alpha)$ B. $(\alpha,0,0)$ C. $(0,0,1)$ D. $(0,\alpha,0)$

gate1993 eigen-value linear-algebra easy multiple-selects

Answer key

6.3.2 Eigen Value: GATE CSE 2002 | Question: 5a



Obtain the eigen values of the matrix

$$A = \begin{bmatrix} 1 & 2 & 34 & 49 \\ 0 & 2 & 43 & 94 \\ 0 & 0 & -2 & 104 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

gatecse-2002 linear-algebra eigen-value normal descriptive

Answer key

6.3.3 Eigen Value: GATE CSE 2005 | Question: 49



What are the eigenvalues of the following 2×2 matrix?

$$\begin{pmatrix} 2 & -1 \\ -4 & 5 \end{pmatrix}$$

- A. -1 and 1 B. 1 and 6 C. 2 and 5 D. 4 and -1

Answer key

6.3.4 Eigen Value: GATE CSE 2007 | Question: 25



Let A be a 4×4 matrix with eigen values $-5, -2, 1, 4$. Which of the following is an eigen value of the matrix

$\begin{bmatrix} A & I \\ I & A \end{bmatrix}$, where I is the 4×4 identity matrix?

- A. -5 B. -7 C. 2 D. 1

Answer key

6.3.5 Eigen Value: GATE CSE 2008 | Question: 28



How many of the following matrices have an eigenvalue 1?

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$

- A. one B. two C. three D. four

Answer key

6.3.6 Eigen Value: GATE CSE 2010 | Question: 29



Consider the following matrix

$$A = \begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

If the eigenvalues of A are 4 and 8, then

- A. $x = 4, y = 10$ B. $x = 5, y = 8$ C. $x = 3, y = 9$ D. $x = -4, y = 10$

Answer key

6.3.7 Eigen Value: GATE CSE 2011 | Question: 40



Consider the matrix as given below.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{bmatrix}$$

Which one of the following options provides the **CORRECT** values of the eigenvalues of the matrix?

- A. 1, 4, 3 B. 3, 7, 3 C. 7, 3, 2 D. 1, 2, 3

Answer key

6.3.8 Eigen Value: GATE CSE 2012 | Question: 11



Let A be the 2×2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

- A. 1024 and -1024 B. $1024\sqrt{2}$ and $-1024\sqrt{2}$
C. $4\sqrt{2}$ and $-4\sqrt{2}$ D. $512\sqrt{2}$ and $-512\sqrt{2}$

Answer key

6.3.9 Eigen Value: GATE CSE 2014 Set 1 | Question: 5



The value of the dot product of the eigenvectors corresponding to any pair of different eigenvalues of a 4×4 symmetric positive definite matrix is _____

gatecse-2014-set1 linear-algebra eigen-value numerical-answers normal

Answer key

6.3.10 Eigen Value: GATE CSE 2014 Set 2 | Question: 47



The product of the non-zero eigenvalues of the matrix is _____

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

gatecse-2014-set2 linear-algebra eigen-value normal numerical-answers

Answer key

6.3.11 Eigen Value: GATE CSE 2014 Set 3 | Question: 4



Which one of the following statements is TRUE about every $n \times n$ matrix with only real eigenvalues?

- A. If the trace of the matrix is positive and the determinant of the matrix is negative, at least one of its eigenvalues is negative.
- B. If the trace of the matrix is positive, all its eigenvalues are positive.
- C. If the determinant of the matrix is positive, all its eigenvalues are positive.
- D. If the product of the trace and determinant of the matrix is positive, all its eigenvalues are positive.

gatecse-2014-set3 linear-algebra eigen-value normal

Answer key

6.3.12 Eigen Value: GATE CSE 2015 Set 1 | Question: 36



Consider the following 2×2 matrix A where two elements are unknown and are marked by a and b . The eigenvalues of this matrix are -1 and 7 . What are the values of a and b ?

$$A = \begin{pmatrix} 1 & 4 \\ b & a \end{pmatrix}$$

- A. $a = 6, b = 4$
- B. $a = 4, b = 6$
- C. $a = 3, b = 5$
- D. $a = 5, b = 3$

gatecse-2015-set1 linear-algebra eigen-value easy

Answer key

6.3.13 Eigen Value: GATE CSE 2015 Set 2 | Question: 5



The larger of the two eigenvalues of the matrix $\begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$ is _____.

gatecse-2015-set2 linear-algebra eigen-value easy numerical-answers

Answer key

6.3.14 Eigen Value: GATE CSE 2015 Set 3 | Question: 15



In the given matrix $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, one of the eigenvalues is 1. The eigenvectors corresponding to the eigenvalue 1 are

- A. $\{a(4, 2, 1) \mid a \neq 0, a \in \mathbb{R}\}$

- B. $\{a(-4, 2, 1) \mid a \neq 0, a \in \mathbb{R}\}$
 C. $\{a(\sqrt{2}, 0, 1) \mid a \neq 0, a \in \mathbb{R}\}$
 D. $\{a(-\sqrt{2}, 0, 1) \mid a \neq 0, a \in \mathbb{R}\}$

gatecse-2015-set3 linear-algebra eigen-value normal

Answer key

6.3.15 Eigen Value: GATE CSE 2016 Set 1 | Question: 05



Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is _____

gatecse-2016-set1 linear-algebra eigen-value numerical-answers normal

Answer key

6.3.16 Eigen Value: GATE CSE 2016 Set 2 | Question: 06



Suppose that the eigenvalues of matrix A are 1, 2, 4. The determinant of $(A^{-1})^T$ is _____.

gatecse-2016-set2 linear-algebra eigen-value normal numerical-answers

Answer key

6.3.17 Eigen Value: GATE CSE 2017 Set 1 | Question: 31



Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2 = 50$. Consider the following statements.

- I. One eigenvalue must be in $[-5, 5]$
- II. The eigenvalue with the largest magnitude must be strictly greater than 5

Which of the above statements about eigenvalues of A is/are necessarily CORRECT?

- A. Both I and II B. I only C. II only D. Neither I nor II

gatecse-2017-set1 linear-algebra eigen-value normal

Answer key

6.3.18 Eigen Value: GATE CSE 2017 Set 2 | Question: 22



Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices.

Then the rank of $P + Q$ is _____.

gatecse-2017-set2 linear-algebra eigen-value numerical-answers

Answer key

6.3.19 Eigen Value: GATE CSE 2017 Set 2 | Question: 52



If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____

gatecse-2017-set2 engineering-mathematics linear-algebra numerical-answers eigen-value

Answer key

6.3.20 Eigen Value: GATE CSE 2018 | Question: 17



Consider a matrix $A = uv^T$ where $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Note that v^T denotes the transpose of v . The largest eigenvalue of A is _____

gatecse-2018 linear-algebra eigen-value normal numerical-answers one-mark

6.3.21 Eigen Value: GATE CSE 2018 | Question: 26



Consider a matrix P whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

Consider the following statements.

- I. P does not have an inverse
- II. P has a repeated eigenvalue
- III. P cannot be diagonalized

Which one of the following options is correct?

- A. Only I and III are necessarily true
- B. Only II is necessarily true
- C. Only I and II are necessarily true
- D. Only II and III are necessarily true

gatecse-2018 linear-algebra matrix eigen-value normal two-marks

6.3.22 Eigen Value: GATE CSE 2019 | Question: 44



Consider the following matrix:

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

The absolute value of the product of Eigen values of R is _____

gatecse-2019 numerical-answers engineering-mathematics linear-algebra eigen-value two-marks

6.3.23 Eigen Value: GATE CSE 2021 Set 1 | Question: 52



Consider the following matrix.

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

The largest eigenvalue of the above matrix is _____.

gatecse-2021-set1 linear-algebra matrix eigen-value numerical-answers two-marks

6.3.24 Eigen Value: GATE CSE 2022 | Question: 43



Which of the following is/are the eigenvector(s) for the matrix given below?

$$\begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$$

A. $\begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

B. $\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$

D. $\begin{pmatrix} 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$

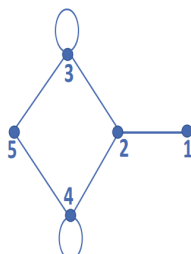
gatecse-2022 linear-algebra eigen-value multiple-selects two-marks

Answer key

6.3.25 Eigen Value: GATE CSE 2023 | Question: 20



Let A be the adjacency matrix of the graph with vertices $\{1, 2, 3, 4, 5\}$.



Let $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, and λ_5 be the five eigenvalues of A . Note that these eigenvalues need not be distinct. The value of $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 =$ _____

gatecse-2023 linear-algebra eigen-value numerical-answers one-mark

Answer key

6.3.26 Eigen Value: GATE CSE 2024 | Set 1 | Question: 2



The product of all eigenvalues of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

- A. -1 B. 0 C. 1 D. 2

gatecse2024-set1 linear-algebra eigen-value one-mark

Answer key

6.3.27 Eigen Value: GATE CSE 2025 | Set 1 | Question: 31



Let A be a 2×2 matrix as given.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What are the eigenvalues of the matrix A^{13} ?

- A. $1, -1$ B. $2\sqrt{2}, -2\sqrt{2}$
C. $4\sqrt{2}, -4\sqrt{2}$ D. $64\sqrt{2}, -64\sqrt{2}$

gatecse2025-set1 linear-algebra matrix eigen-value easy two-marks

Answer key

6.3.28 Eigen Value: GATE DA 2025 | Question: 18



Let $A = I_n + xx^\top$, where I_n is the $n \times n$ identity matrix and $x \in \mathbb{R}^n, x^\top x = 1$. Which of the following options is/are correct?

- A. Rank of A is n B. A is invertible
C. 0 is an eigenvalue of A D. A^{-1} has a negative eigenvalue

gateda-2025 linear-algebra matrix eigen-value multiple-selects one-mark

Answer key

6.3.29 Eigen Value: GATE DS&AI 2024 | Question: 3



Consider the matrix $M = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$.

Which ONE of the following statements is TRUE?

- A. The eigenvalues of M are non-negative and real.
- B. The eigenvalues of M are complex conjugate pairs.
- C. One eigenvalue of M is positive and real, and another eigenvalue of M is zero.
- D. One eigenvalue of M is non-negative and real, and another eigenvalue of M is negative and real.

gate-ds-ai-2024 eigen-value linear-algebra one-mark

Answer key

6.3.30 Eigen Value: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 17



For matrix $H = \begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$, one of the eigenvalues is 5. Then, the other eigenvalue is

- A. 12
- B. 10
- C. 8
- D. 6

gateda-sample-paper-2024 linear-algebra eigen-value easy

Answer key

6.3.31 Eigen Value: GATE IT 2006 | Question: 26



What are the eigenvalues of the matrix P given below

$$P = \begin{pmatrix} a & 1 & 0 \\ 1 & a & 1 \\ 0 & 1 & a \end{pmatrix}$$

- A. $a, a - \sqrt{2}, a + \sqrt{2}$
- B. a, a, a
- C. $0, a, 2a$
- D. $-a, 2a, 2a$

gateit-2006 linear-algebra eigen-value normal

Answer key

6.3.32 Eigen Value: GATE IT 2007 | Question: 2



Let A be the matrix $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. What is the maximum value of $x^T A x$ where the maximum is taken over all x that are the unit eigenvectors of A ?

- A. 5
- B. $\frac{(5+\sqrt{5})}{2}$
- C. 3
- D. $\frac{(5-\sqrt{5})}{2}$

gateit-2007 linear-algebra eigen-value normal

Answer key

6.3.33 Eigen Value: GATE-2014 EC



Which one of the following statements is NOT true for a square matrix A ?

- a. If A is real symmetric, the eigen values of A are the diagonal elements of it.
- b. If all the principal minors of A are positive, all the eigen values of A are also positive.

My question is what is "principal minors of A " ?

linear-algebra gate2014-ec-1 eigen-value

Answer key

6.4

Gaussian Elimination (1)

6.4.1 Gaussian Elimination: GATE DA 2025 | Question: 2



The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order

- A. $O(n)$ B. $O(n^2)$ C. $O(n^3)$ D. $O(n^4)$

gateda-2025 linear-algebra matrix gaussian-elimination one-mark

Answer key

6.5

Lu Decomposition (1)

6.5.1 Lu Decomposition: GATE CSE 2025 | Set 2 | Question: 34



Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3 \times 3}$ and $Q \in \mathbb{R}^{3 \times 1}$. Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Which of the following statement(s) is/are TRUE?

- A. The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
 B. If P is invertible, then both L and U are invertible.
 C. If P is singular, then at least one of the diagonal elements of U is zero.
 D. If P is symmetric, then both L and U are symmetric.

gatecse2025-set2 linear-algebra system-of-equations matrix lu-decomposition multiple-selects two-marks

Answer key

6.6

Matrix (22)

6.6.1 Matrix: GATE CSE 1987 | Question: 1-xxiii



A square matrix is singular whenever

- A. The rows are linearly independent B. The columns are linearly independent
 C. The row are linearly dependent D. None of the above

gate1987 linear-algebra matrix

Answer key

6.6.2 Matrix: GATE CSE 1988 | Question: 16i



Assume that the matrix A given below, has factorization of the form $LU = PA$, where L is lower-triangular with all diagonal elements equal to 1, U is upper-triangular, and P is a permutation matrix. For

$$A = \begin{bmatrix} 2 & 5 & 9 \\ 4 & 6 & 5 \\ 8 & 2 & 3 \end{bmatrix}$$

Compute L, U , and P using Gaussian elimination with partial pivoting.

gate1988 normal descriptive linear-algebra matrix

Answer key

6.6.3 Matrix: GATE CSE 1993 | Question: 02.7



If $A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & i & i \\ 0 & 0 & 0 & -i \end{pmatrix}$ the matrix A^4 , calculated by the use of Cayley-Hamilton theorem or otherwise, is

gate1993 linear-algebra normal matrix fill-in-the-blanks

Answer key

6.6.4 Matrix: GATE CSE 1994 | Question: 1.2



Let A and B be real symmetric matrices of size $n \times n$. Then which one of the following is true?

- A. $AA' = I$ B. $A = A^{-1}$ C. $AB = BA$ D. $(AB)' = BA$

gate1994 linear-algebra normal matrix

Answer key

6.6.5 Matrix: GATE CSE 1994 | Question: 3.12



Find the inverse of the matrix $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

gate1994 linear-algebra matrix easy descriptive

Answer key

6.6.6 Matrix: GATE CSE 1996 | Question: 10



Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ be two matrices such that $AB = I$. Let $C = A \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $CD = I$. Express the elements of D in terms of the elements of B .

gate1996 linear-algebra matrix normal descriptive

Answer key

6.6.7 Matrix: GATE CSE 1996 | Question: 2.6



The matrices $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ and $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under multiplication

- A. if $a = b$ or $\theta = n\pi, n$ an integer B. always
C. never D. if $a \cos \theta = b \sin \theta$

gate1996 linear-algebra normal matrix

Answer key

6.6.8 Matrix: GATE CSE 1997 | Question: 4.2



Let $A = (a_{ij})$ be an n -rowed square matrix and I_{12} be the matrix obtained by interchanging the first and second rows of the n -rowed Identity matrix. Then AI_{12} is such that its first

- A. Row is the same as its second row B. Row is the same as the second row of A
C. Column is the same as the second column of A D. Row is all zero

gate1997 linear-algebra easy matrix

Answer key

6.6.9 Matrix: GATE CSE 1998 | Question: 2.2



Consider the following determinant $\Delta = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$

Which of the following is a factor of Δ ?

- A. $a + b$ B. $a - b$ C. $a + b + c$ D. abc

gate1998 linear-algebra matrix normal

Answer key 

6.6.10 Matrix: GATE CSE 2001 | Question: 1.1



Consider the following statements:

- S1: The sum of two singular $n \times n$ matrices may be non-singular
- S2: The sum of two $n \times n$ non-singular matrices may be singular

Which one of the following statements is correct?

- A. S1 and S2 both are true B. S1 is true, S2 is false
C. S1 is false, S2 is true D. S1 and S2 both are false

gatecse-2001 linear-algebra normal matrix

Answer key 

6.6.11 Matrix: GATE CSE 2004 | Question: 26



The number of different $n \times n$ symmetric matrices with each element being either 0 or 1 is: (Note: power $(2, X)$ is same as 2^X)

- A. power $(2, n)$ B. power $(2, n^2)$
C. power $\left(2, \frac{(n^2+n)}{2}\right)$ D. power $\left(2, \frac{(n^2-n)}{2}\right)$

gatecse-2004 linear-algebra normal matrix

Answer key 

6.6.12 Matrix: GATE CSE 2004 | Question: 27



Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = I$, then B^{-1} is

- A. $D^{-1}C^{-1}A^{-1}$ B. CDA
C. ADC D. Does not necessarily exist

gatecse-2004 linear-algebra normal matrix

Answer key 

6.6.13 Matrix: GATE CSE 2004 | Question: 76



In an $M \times N$ matrix all non-zero entries are covered in a rows and b columns. Then the maximum number of non-zero entries, such that no two are on the same row or column, is

- A. $\leq a + b$ B. $\leq \max(a, b)$
C. $\leq \min(M - a, N - b)$ D. $\leq \min(a, b)$

gatecse-2004 linear-algebra normal matrix

Answer key 

6.6.14 Matrix: GATE CSE 2006 | Question: 23



F is an $n \times n$ real matrix. b is an $n \times 1$ real vector. Suppose there are two $n \times 1$ vectors, u and v such that, $u \neq v$ and $Fu = b, Fv = b$. Which one of the following statements is false?

- A. Determinant of F is zero.
B. There are an infinite number of solutions to $Fx = b$
C. There is an $x \neq 0$ such that $Fx = 0$
D. F must have two identical rows

gatecse-2006 linear-algebra normal matrix

Answer key 

6.6.15 Matrix: GATE CSE 2015 Set 1 | Question: 18



In the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 \\ 4 & 9 \end{bmatrix}$, if the diagonal elements of U are both 1, then the lower diagonal entry l_{22} of L is _____.

gatecse-2015-set1 linear-algebra matrix numerical-answers

Answer key

6.6.16 Matrix: GATE CSE 2015 Set 2 | Question: 27



Perform the following operations on the matrix $\begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$

- Add the third row to the second row
- Subtract the third column from the first column.

The determinant of the resultant matrix is _____.

gatecse-2015-set2 linear-algebra matrix easy numerical-answers

Answer key

6.6.17 Matrix: GATE CSE 2022 | Question: 10



Consider the following two statements with respect to the matrices $A_{m \times n}$, $B_{n \times m}$, $C_{n \times n}$ and $D_{n \times n}$.

Statement 1 : $tr(AB) = tr(BA)$

Statement 2 : $tr(CD) = tr(DC)$

where $tr()$ represents the trace of a matrix. Which one of the following holds?

- Statement 1 is correct and Statement 2 is wrong.
- Statement 1 is wrong and Statement 2 is correct.
- Both Statement 1 and Statement 2 are correct.
- Both Statement 1 and Statement 2 are wrong.

gatecse-2022 linear-algebra matrix one-mark

Answer key

6.6.18 Matrix: GATE CSE 2025 | Set 2 | Question: 1



If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, then which ONE of the following is A^8 ?

- | | |
|---|---|
| A. $\begin{pmatrix} 25 & 0 \\ 0 & 25 \end{pmatrix}$ | B. $\begin{pmatrix} 125 & 0 \\ 0 & 125 \end{pmatrix}$ |
| C. $\begin{pmatrix} 625 & 0 \\ 0 & 625 \end{pmatrix}$ | D. $\begin{pmatrix} 3125 & 0 \\ 0 & 3125 \end{pmatrix}$ |

gatecse2025-set2 linear-algebra matrix easy one-mark

Answer key

6.6.19 Matrix: GATE DA 2025 | Question: 27



Let $A \in \mathbb{R}^{n \times n}$ be such that $A^3 = A$. Which one of the following statements is ALWAYS correct?

- | | |
|---|-------------------------------------|
| A. A is invertible | B. Determinant of A is 0 |
| C. The sum of the diagonal elements of A is 1 | D. A and A^2 have the same rank |

gateda-2025 linear-algebra matrix two-marks

Answer key

6.6.20 Matrix: GATE DA 2025 | Question: 42



An $n \times n$ matrix A with real entries satisfies the property: $\|Ax\|^2 = \|x\|^2$, for all $x \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the

Euclidean norm. Which of the following statements is/are ALWAYS correct?

- A. A must be orthogonal
B. $A = I$, where I denotes the identity matrix, is the only solution
C. The eigenvalues of A are either $+1$ or -1
D. A has full rank

gateda-2025 linear-algebra matrix multiple-selects two-marks

Answer key

6.6.21 Matrix: GATE IT 2004 | Question: 36



If matrix $X = \begin{bmatrix} a & 1 \\ -a^2 + a - 1 & 1 - a \end{bmatrix}$ and $X^2 - X + I = O$ (I is the identity matrix and O is the zero matrix), then the inverse of X is

- A. $\begin{bmatrix} 1-a & -1 \\ a^2 & a \end{bmatrix}$
B. $\begin{bmatrix} 1-a & -1 \\ a^2 - a + 1 & a \end{bmatrix}$
C. $\begin{bmatrix} -a & 1 \\ -a^2 + a - 1 & 1 - a \end{bmatrix}$
D. $\begin{bmatrix} a^2 - a + 1 & a \\ 1 & 1 - a \end{bmatrix}$

gateit-2004 linear-algebra matrix normal

Answer key

6.6.22 Matrix: GATE IT 2008 | Question: 29



If M is a square matrix with a zero determinant, which of the following assertion (s) is (are) correct?

- S1:** Each row of M can be represented as a linear combination of the other rows
S2: Each column of M can be represented as a linear combination of the other columns
S3: $MX = 0$ has a nontrivial solution
S4: M has an inverse

- A. $S3$ and $S2$
B. $S1$ and $S4$
C. $S1$ and $S3$
D. $S1, S2$ and $S3$

gateit-2008 linear-algebra normal matrix

Answer key

6.7 Orthonormality (2)

6.7.1 Orthonormality: GATE DA 2025 | Question: 15



Which of the following statements is/are correct?

- A. $\mathbb{R}^n$ has a unique set of orthonormal basis vectors
B. $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors
C. Linearly independent vectors in $\mathbb{R}^n$ are orthonormal
D. Orthonormal vectors in $\mathbb{R}^n$ are linearly independent

gateda-2025 linear-algebra vector-space orthonormality multiple-selects one-mark

Answer key

6.7.2 Orthonormality: GATE DA 2025 | Question: 40



Let x_1, x_2, x_3, x_4, x_5 be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $A = x_1 x_1^\top + \dots + x_5 x_5^\top$. Which of the following statements is/are correct?

- A. Singular values of A are also its eigenvalues
B. Singular values of A are either 0 or 1
C. Determinant of A is 1
D. A is invertible

gateda-2025 linear-algebra orthonormality vector-space matrix multiple-selects two-marks

Answer key

6.8 Rank of Matrix (6)

6.8.1 Rank of Matrix: GATE CSE 1994 | Question: 1.9



The rank of matrix $\begin{bmatrix} 0 & 0 & -3 \\ 9 & 3 & 5 \\ 3 & 1 & 1 \end{bmatrix}$ is:

- A. 0 B. 1 C. 2 D. 3

gate1994 linear-algebra matrix rank-of-matrix easy

Answer key 

6.8.2 Rank of Matrix: GATE CSE 1995 | Question: 1.24



The rank of the following $(n+1) \times (n+1)$ matrix, where a is a real number is

$$\begin{bmatrix} 1 & a & a^2 & \dots & a^n \\ 1 & a & a^2 & \dots & a^n \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a & a^2 & \dots & a^n \end{bmatrix}$$

- A. 1 B. 2
C. n D. Depends on the value of a

gate1995 linear-algebra matrix normal rank-of-matrix

Answer key 

6.8.3 Rank of Matrix: GATE CSE 1998 | Question: 2.1



The rank of the matrix given below is:

$$\begin{bmatrix} 1 & 4 & 8 & 7 \\ 0 & 0 & 3 & 0 \\ 4 & 2 & 3 & 1 \\ 3 & 12 & 24 & 21 \end{bmatrix}$$

- A. 3 B. 1 C. 2 D. 4

gate1998 linear-algebra matrix normal rank-of-matrix

Answer key 

6.8.4 Rank of Matrix: GATE CSE 2002 | Question: 1.1



The rank of the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ is

- A. 4 B. 2 C. 1 D. 0

gatecse-2002 linear-algebra easy rank-of-matrix

Answer key 

6.8.5 Rank of Matrix: GATE CSE 2020 | Question: 27



Let A and B be two $n \times n$ matrices over real numbers. Let $\text{rank}(M)$ and $\det(M)$ denote the rank and determinant of a matrix M , respectively. Consider the following statements.

- I. $\text{rank}(AB) = \text{rank}(A)\text{rank}(B)$
- II. $\det(AB) = \det(A)\det(B)$
- III. $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$
- IV. $\det(A+B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

- A. I and II only B. I and IV only C. II and III only D. III and IV only

Answer key

6.8.6 Rank of Matrix: GATE CSE 2021 Set 2 | Question: 24



Suppose that P is a 4×5 matrix such that every solution of the equation $Px=0$ is a scalar multiple of $[2 \ 5 \ 4 \ 3 \ 1]^T$. The rank of P is _____

gatecse-2021-set2 numerical-answers linear-algebra matrix rank-of-matrix one-mark

Answer key

6.9

Singular Value Decomposition (1)

6.9.1 Singular Value Decomposition: GATE DS&AI 2024 | Question: 51



Let $u = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$, and let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$ be the singular values of the matrix $M = uu^T$ (where u^T is the transpose of u). The value of $\sum_{i=1}^5 \sigma_i$ is _____

gate-ds-ai-2024 linear-algebra singular-value-decomposition numerical-answers two-marks

Answer key

6.10

Subspace (1)

6.10.1 Subspace: GATE DS&AI 2024 | Question: 37



Select all choices that are subspaces of $\mathbb{R}^3$.

Note: $\mathbb{R}$ denotes the set of real numbers.

- A. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
- B. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
- C. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0 \right\}$
- D. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 + 4 = 0 \right\}$

gate-ds-ai-2024 linear-algebra vector-space subspace multiple-selects two-marks

Answer key

6.11

System of Equations (17)

6.11.1 System of Equations: GATE CSE 1996 | Question: 1.7



Let $Ax = b$ be a system of linear equations where A is an $m \times n$ matrix and b is a $m \times 1$ column vector and X is an $n \times 1$ column vector of unknowns. Which of the following is false?

- A. The system has a solution if and only if, both A and the augmented matrix $[Ab]$ have the same rank.
- B. If $m < n$ and b is the zero vector, then the system has infinitely many solutions.
- C. If $m = n$ and b is a non-zero vector, then the system has a unique solution.
- D. The system will have only a trivial solution when $m = n$, b is the zero vector and $\text{rank}(A) = n$.

gate1996 linear-algebra system-of-equations normal

Answer key

6.11.2 System of Equations: GATE CSE 1998 | Question: 1.2



Consider the following set of equations

- $x + 2y = 5$
- $4x + 8y = 12$
- $3x + 6y + 3z = 15$

This set

- A. has unique solution
 B. has no solution
 C. has finite number of solutions
 D. has infinite number of solutions

gate1998 linear-algebra system-of-equations easy

Answer key

6.11.3 System of Equations: GATE CSE 1998 | Question: 9



Derive the expressions for the number of operations required to solve a system of linear equations in n unknowns using the Gaussian Elimination Method. Assume that one operation refers to a multiplication followed by an addition.

gate1998 linear-algebra system-of-equations descriptive

Answer key

6.11.4 System of Equations: GATE CSE 2003 | Question: 41



Consider the following system of linear equations

$$\begin{pmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha \\ 5 \\ 7 \end{pmatrix}$$

Notice that the second and the third columns of the coefficient matrix are linearly dependent. For how many values of α , does this system of equations have infinitely many solutions?

- A. 0
 B. 1
 C. 2
 D. 3

gatecse-2003 linear-algebra system-of-equations normal

Answer key

6.11.5 System of Equations: GATE CSE 2004 | Question: 71



How many solutions does the following system of linear equations have?

- $-x + 5y = -1$
- $x - y = 2$
- $x + 3y = 3$

- A. infinitely many
 B. two distinct solutions
 C. unique
 D. none

gatecse-2004 linear-algebra system-of-equations normal

Answer key

6.11.6 System of Equations: GATE CSE 2005 | Question: 48



Consider the following system of linear equations :

$$2x_1 - x_2 + 3x_3 = 1$$

$$3x_1 + 2x_2 + 5x_3 = 2$$

$$-x_1 + 4x_2 + x_3 = 3$$

The system of equations has

- A. no solution
 C. more than one but a finite number of solutions
 B. a unique solution
 D. an infinite number of solutions

gatecse-2005 linear-algebra system-of-equations normal

Answer key

6.11.7 System of Equations: GATE CSE 2008 | Question: 3



The following system of equations

- $x_1 + x_2 + 2x_3 = 1$
- $x_1 + 2x_2 + 3x_3 = 2$
- $x_1 + 4x_2 + \alpha x_3 = 4$

has a unique solution. The only possible value(s) for α is/are

- A. 0
 B. either 0 or 1
 C. one of 0, 1, or -1
 D. any real number

gatecse-2008 easy linear-algebra system-of-equations

Answer key

6.11.8 System of Equations: GATE CSE 2014 Set 1 | Question: 4



Consider the following system of equations:

- $3x + 2y = 1$
- $4x + 7z = 1$
- $x + y + z = 3$
- $x - 2y + 7z = 0$

The number of solutions for this system is _____

gatecse-2014-set1 linear-algebra system-of-equations numerical-answers normal

Answer key

6.11.9 System of Equations: GATE CSE 2015 Set 3 | Question: 33



If the following system has non-trivial solution,

- $px + qy + rz = 0$
- $qx + ry + pz = 0$
- $rx + py + qz = 0,$

then which one of the following options is TRUE?

- A. $p - q + r = 0$ or $p = q = -r$
 C. $p + q + r = 0$ or $p = q = r$
 B. $p + q - r = 0$ or $p = -q = r$
 D. $p - q + r = 0$ or $p = -q = -r$

gatecse-2015-set3 linear-algebra system-of-equations normal

Answer key

6.11.10 System of Equations: GATE CSE 2016 Set 2 | Question: 04



Consider the systems, each consisting of m linear equations in n variables.

- If $m < n$, then all such systems have a solution.
- If $m > n$, then none of these systems has a solution.
- If $m = n$, then there exists a system which has a solution.

Which one of the following is **CORRECT**?

- A. I, II and III are true.
 B. Only II and III are true.

C. Only III is true.

D. None of them is true.

gatecse-2016-set2 linear-algebra system-of-equations normal

Answer key

6.11.11 System of Equations: GATE CSE 2017 Set 1 | Question: 3



Let $c_1, \dots, c_n$ be scalars, not all zero, such that $\sum_{i=1}^n c_i a_i = 0$ where a_i are column vectors in R^n .

Consider the set of linear equations

$$Ax = b$$

where $A = [a_1, \dots, a_n]$ and $b = \sum_{i=1}^n a_i$. The set of equations has

- A. a unique solution at $x = J_n$ where J_n denotes a n -dimensional vector of all 1.
- B. no solution
- C. infinitely many solutions
- D. finitely many solutions

gatecse-2017-set1 linear-algebra system-of-equations normal

Answer key

6.11.12 System of Equations: GATE CSE 2022 | Question: 35



Consider solving the following system of simultaneous equations using LU decomposition.

$$x_1 + x_2 - 2x_3 = 4$$

$$x_1 + 3x_2 - x_3 = 7$$

$$2x_1 + x_2 - 5x_3 = 7$$

where L and U are denoted as

$$L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}$$

Which one of the following is the correct combination of values for L_{32} , U_{33} , and x_1 ?

- A. $L_{32} = 2, U_{33} = -\frac{1}{2}, x_1 = -1$
- B. $L_{32} = 2, U_{33} = 2, x_1 = -1$
- C. $L_{32} = -\frac{1}{2}, U_{33} = 2, x_1 = 0$
- D. $L_{32} = -\frac{1}{2}, U_{33} = -\frac{1}{2}, x_1 = 0$

gatecse-2022 linear-algebra matrix system-of-equations two-marks

Answer key

6.11.13 System of Equations: GATE CSE 2024 | Set 1 | Question: 39



Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

- A. There exist at least $m - n$ linearly independent solutions to this system
- B. There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors
- C. There exists a non-zero solution in which at least $m - n$ variables are 0
- D. There exists a solution in which at least n variables are non-zero

gatecse2024-set1 multiple-selects linear-algebra system-of-equations two-marks

Answer key

6.11.14 System of Equations: GATE CSE 2025 | Set 1 | Question: 13



Consider the given system of linear equations for variables x and y , where k is a real-valued constant. Which of the following option(s) is/are CORRECT?

$$x + ky = 1$$

$$kx + y = -1$$

- A. There is exactly one value of k for which the above system of equations has no solution.
- B. There exist an infinite number of values of k for which the system of equations has no solution.
- C. There exists exactly one value of k for which the system of equations has exactly one solution.
- D. There exists exactly one value of k for which the system of equations has an infinite number of solutions.

gatecse2025-set1 linear-algebra system-of-equations multiple-selects one-mark

Answer key

6.11.15 System of Equations: GATE DA 2025 | Question: 3



The sum of the elements in each row of $A \in \mathbb{R}^{n \times n}$ is 1. If $B = A^3 - 2A^2 + A$, which one of the following statements is correct (for $x \in \mathbb{R}^n$)?

- A. The equation $Bx = 0$ has no solution
- B. The equation $Bx = 0$ has exactly two solutions
- C. The equation $Bx = 0$ has infinitely many solutions
- D. The equation $Bx = 0$ has a unique solution

gateda-2025 linear-algebra system-of-equations one-mark

Answer key

6.11.16 System of Equations: GATE DS&AI 2024 | Question: 38



Which of the following statements is/are TRUE?

Note: $\mathbb{R}$ denotes the set of real numbers.

- A. There exist $M \in \mathbb{R}^{3 \times 3}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has a unique solution and $Mx = q$ has infinite solutions.
- B. There exist $M \in \mathbb{R}^{3 \times 3}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has no solutions and $Mx = q$ has infinite solutions.
- C. There exist $M \in \mathbb{R}^{2 \times 3}$, $p \in \mathbb{R}^2$, and $q \in \mathbb{R}^2$ such that $Mx = p$ has a unique solution and $Mx = q$ has infinite solutions.
- D. There exist $M \in \mathbb{R}^{3 \times 2}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has a unique solution and $Mx = q$ has no solutions.

linear-algebra system-of-equations gate-ds-ai-2024 multiple-selects two-marks

Answer key

6.11.17 System of Equations: GATE IT 2004 | Question: 6



What values of x , y and z satisfy the following system of linear equations?

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 12 \end{bmatrix}$$

- A. $x = 6, y = 3, z = 2$
- B. $x = 12, y = 3, z = -4$
- C. $x = 6, y = 6, z = -4$
- D. $x = 12, y = -3, z = 0$

gateit-2004 linear-algebra system-of-equations easy

Answer key

6.12

Vector Space (6)

6.12.1 Vector Space: GATE CSE 1995 | Question: 2.13



A unit vector perpendicular to both the vectors $a = 2i - 3j + k$ and $b = i + j - 2k$ is:

- A. $\frac{1}{\sqrt{3}}(i + j + k)$
- B. $\frac{1}{3}(i + j - k)$
- C. $\frac{1}{3}(i - j - k)$
- D. $\frac{1}{\sqrt{3}}(i + j - k)$

gate1995 linear-algebra normal vector-space

Answer key

6.12.2 Vector Space: GATE CSE 2007 | Question: 27



Consider the set of (column) vectors defined by

$$X = \left\{ x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0, \text{ where } x^T = [x_1, x_2, x_3]^T \right\}$$

.Which of the following is TRUE?

- A. $\left\{ [1, -1, 0]^T, [1, 0, -1]^T \right\}$ is a basis for the subspace X .
- B. $\left\{ [1, -1, 0]^T, [1, 0, -1]^T \right\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X .
- C. X is not a subspace of $\mathbb{R}^3$.
- D. None of the above

gatecse-2007 linear-algebra normal vector-space

Answer key

6.12.3 Vector Space: GATE CSE 2014 Set 3 | Question: 5



If V_1 and V_2 are 4-dimensional subspaces of a 6-dimensional vector space V , then the smallest possible dimension of $V_1 \cap V_2$ is _____.

gatecse-2014-set3 linear-algebra vector-space normal numerical-answers

Answer key

6.12.4 Vector Space: GATE CSE 2017 Set 1 | Question: 30



Let u and v be two vectors in $\mathbb{R}^2$ whose Euclidean norms satisfy $\|u\| = 2\|v\|$. What is the value of α such that $w = u + \alpha v$ bisects the angle between u and v ?

- A. 2
- B. $\frac{1}{2}$
- C. 1
- D. $-\frac{1}{2}$

gatecse-2017-set1 linear-algebra normal vector-space

Answer key

6.12.5 Vector Space: GATE DA 2025 | Question: 28



Let $\{x_1, x_2, \dots, x_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i, j) -th element of matrix $A \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = x_i^T x_j, 1 \leq i, j \leq n$. Which one of the following statements is correct?

- A. A is invertible
- B. 0 is a singular value of A
- C. Determinant of A is 0
- D. $z^T A z = 0$ for some non-zero $z \in \mathbb{R}^n$

gateda-2025 linear-algebra vector-space matrix two-marks

Answer key

6.12.6 Vector Space: GATE DS&AI 2024 | Question: 39



Let $\mathbb{R}$ be the set of real numbers, U be a subspace of $\mathbb{R}^3$ and $M \in \mathbb{R}^{3 \times 3}$ be the matrix corresponding to the projection on to the subspace U .

Which of the following statements is/are TRUE?

- A. If U is a 1-dimensional subspace of $\mathbb{R}^3$, then the null space of M is a 1-dimensional subspace.
- B. If U is a 2-dimensional subspace of $\mathbb{R}^3$, then the null space of M is a 1-dimensional subspace.
- C. $M^2 = M$
- D. $M^3 = M$

gate-ds-ai-2024 linear-algebra vector-space multiple-selects two-marks

Answer key

Answer Keys

6.1.1	A
6.2.5	D
6.2.10	D
6.3.4	C
6.3.9	0
6.3.14	B
6.3.19	5
6.3.24	A;C;D
6.3.29	B
6.4.1	B
6.6.4	D
6.6.9	B
6.6.14	D
6.6.19	D
6.7.2	A;B
6.8.5	C
6.11.2	B
6.11.7	X
6.11.12	D
6.11.17	C
6.12.5	A

6.2.1	B
6.2.6	B
6.2.11	A
6.3.5	A
6.3.10	6
6.3.15	15
6.3.20	3
6.3.25	2
6.3.30	B
6.5.1	A;B;C
6.6.5	N/A
6.6.10	A
6.6.15	5
6.6.20	A;D
6.8.1	C
6.8.6	4 : 4
6.11.3	N/A
6.11.8	1
6.11.13	A
6.12.1	A
6.12.6	B;C;D

6.2.2	A
6.2.7	A;B
6.3.1	B;D
6.3.6	D
6.3.11	A
6.3.16	0.125
6.3.21	D
6.3.26	B
6.3.31	A
6.6.1	C
6.6.6	N/A
6.6.11	C
6.6.16	0
6.6.21	B
6.8.2	A
6.9.1	55
6.11.4	B
6.11.9	C
6.11.14	A;D
6.12.2	A

6.2.3	A
6.2.8	A
6.3.2	N/A
6.3.7	A
6.3.12	D
6.3.17	B
6.3.22	12
6.3.27	D
6.3.32	B
6.6.2	N/A
6.6.7	A
6.6.12	B
6.6.17	C
6.6.22	D
6.8.3	A
6.10.1	A;C
6.11.5	C
6.11.10	C
6.11.15	C
6.12.3	2

6.2.4	0
6.2.9	0
6.3.3	B
6.3.8	D
6.3.13	6
6.3.18	2
6.3.23	3 : 3
6.3.28	A;B
6.3.33	Q-Q
6.6.3	N/A
6.6.8	C
6.6.13	D
6.6.18	C
6.7.1	B;D
6.8.4	C
6.11.1	C
6.11.6	B
6.11.11	C
6.11.16	B;D
6.12.4	A